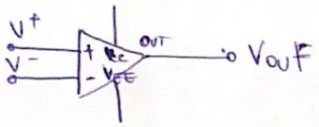
DEEA

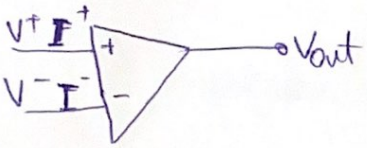
– Curs 8 decembrie 2022. –

AMPLIFICATOARE OPERAȚIONALE

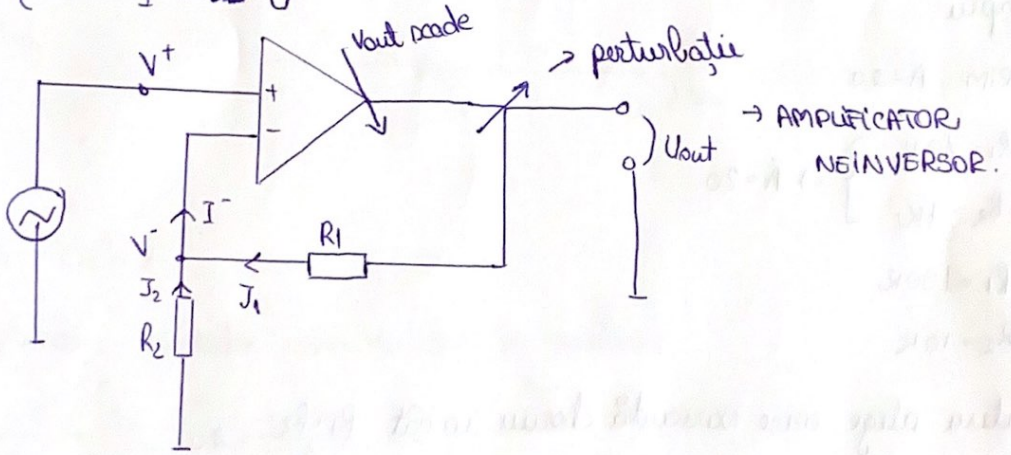


$V_{out} = A(V^+ - V^-)$

MODEL A.O. IDEAL



$$\begin{cases} V_{out} = A(V^+ - V^-) \\ A \to \infty \quad (A = \infty) \\ I^+ = I^- = 0 \end{cases}$$



→ pentru amplificatoare ⇒ METODA POTENȚIALELOR LA NODURI

$V^- = \frac{V_{out}}{A}$

→ Re

$V_{out} = A(V^+ - V^-)$

$V^+ = U_i$

$V^- =$

$I^- + I_2 + I_1 = 0$

$0 + \frac{(0 - V')}{R_2} + \frac{V_{out} - V^-}{R_1} = 0$

$V^-\left(\frac{1}{R_1} + \frac{1}{R_2}\right) = \frac{V_{out}}{R_1}$

$V^- = \frac{V_{out}}{R_1} \cdot \frac{R_1 R_2}{R_1 + R_2} = V_{out} \cdot \frac{R_2}{R_1 + R_2}$

$V_{out} = A\left(U_i - V_{out} \frac{R_2}{R_1 + R_2}\right)$

$V_{out} = \frac{A U_i}{1 + A \cdot \frac{R_2}{R_1 + R_2}} \overset{A \to \infty}{=\!=} \frac{U_i (R_1 + R_2)}{R_2}$.

Exemplu:

DORIM $A = 20$

I: $R_1 = 19K$, $R_2 = 1K$ } ⇒ $A = 20$.

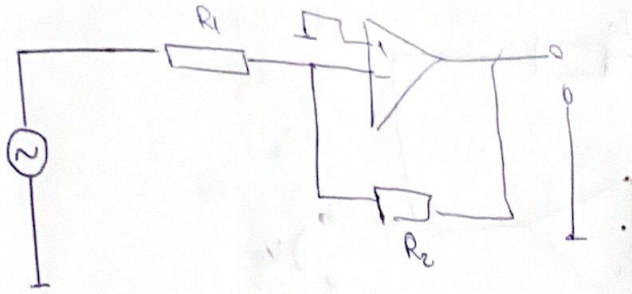
II: $R_1 = 190K$

$R_2 = 10K$

→ Putem alege orice variantă dorim încât $\frac{R_1 + R_2}{R_2} = 20$.

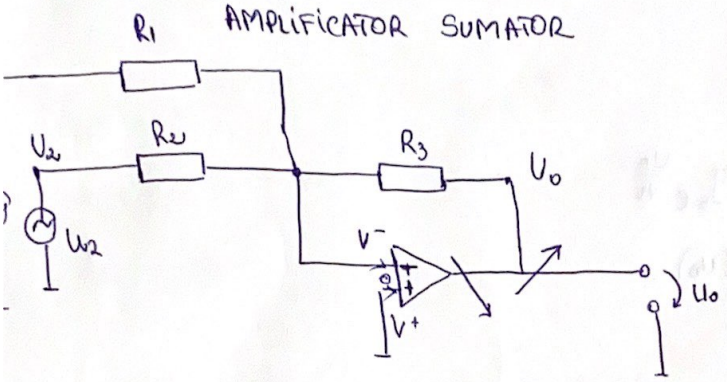
$$= \frac{V_{out}}{A} = \frac{U_i}{A} \cdot \frac{R_1+R_2}{R_2} \overset{A\to\infty}{=} 0$$

→ Reacție negativă (R.N.) ⇒ $V^+ = V^-$



↳ amplificator inversor

AMPLIFICATOR SUMATOR



R.N. ⇒ $V^+ = V^-$

→ Aplicăm metoda potențialelor la noduri

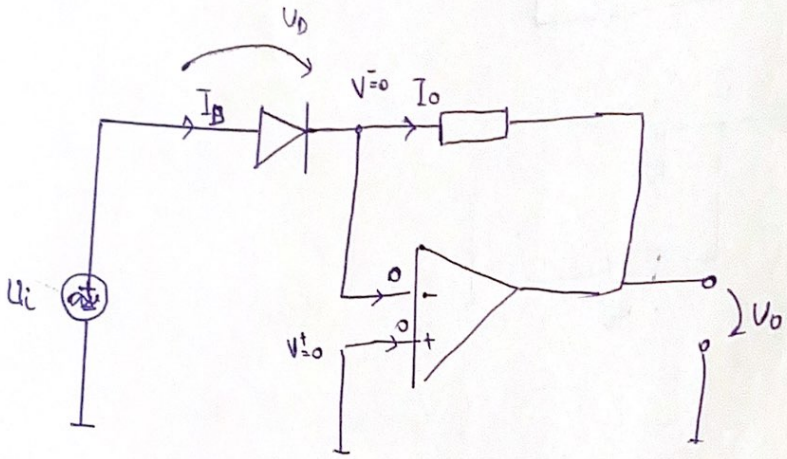
$$\frac{U_1 - V^-}{R_1} + \frac{U_2 - V^+}{R_2} + \frac{U_0 - V^-}{R_3} = 0$$

$$\frac{U_1}{R_1} + \frac{U_2}{R_2} = -\frac{U_0}{R_3}$$

$$U_0 = \left(\frac{U_1}{R_1} + \frac{U_2}{R_2}\right) \cdot (-R_3)$$

ALEGEM $R_1 = R_2 = R_3$

$U_0 = -(U_1 + U_2)$

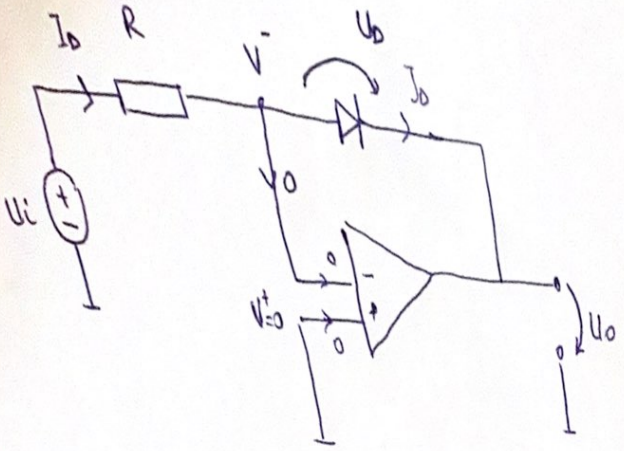


$$\frac{V^- - U_0}{R} = I_D$$

$U^- = V^- = U_D \Rightarrow U_D = U_i$

$$I_D = I_S\left(e^{\frac{U_D}{V_t}} - 1\right) \simeq I_S e^{\frac{U_D}{V_t}}$$

$$U_0 = R I_S \cdot e^{\frac{U_D}{V_T}} \approx \exp(U_D)$$



$V^+ = 0$

R.N. $\Rightarrow V^+ = V^- \Rightarrow V^- = 0$

$$I_D = \frac{U_i - V^+}{R} = \frac{U_i}{R}$$

$$\left.\begin{aligned} I_D &= I_S \cdot e^{\frac{U_D}{V_T}} \\ U_D &= V^- - U_o = -U_o \end{aligned}\right\} = I_S \cdot e^{-\frac{U_o}{V_T}}$$

$$-\frac{U_o}{V_T} = \ln \frac{I_D}{I_S} \Rightarrow$$

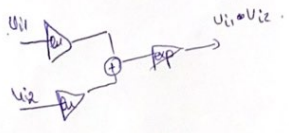
$$U_o = -V_T \ln \frac{I_D}{I_S} = -V_T \ln \frac{U_i}{I_S R} \simeq \ln U_i$$

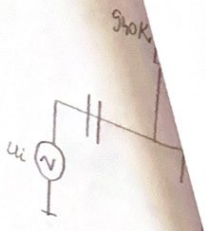
AMPLIFICATOR EXPONENȚIAL. ↗

$$U_{i1} \cdot U_{i2} = \exp(\ln U_{i1} + \ln U_{i2})$$

\+ —

ln exp.





$\frac{I_C}{\beta} \cdot R_1 + U_{BE} = V_{CC} \Rightarrow$

$U_{CE} = V_{CC} - R_2 I_C = 5V >$

$r_i = \frac{?}{\beta I_C} \approx$